

A two-piece construction for the ($2\alpha, \alpha, 2\beta$) family

Denis Paliy · 29 September 2026

DRAFT FOR DISCUSSION

Idea. Glue a known triquadratic tiling to one quadratically subdivided similar copy of the tile. The common side matches exactly, and the two angles at one endpoint add to π . This gives every scale permitted by the elementary boundary-length condition for the specified rational tile and target shape.

Status. Developed with assistance from ChatGPT and internally checked. No external review or claim of priority is made. This result concerns one target-shape family; it does not solve Erdős problem 634 in full.

Theorem

Let $0 < u < v$ be coprime integers and set

$$a = uv, \quad b = v^2 - u^2, \quad c = v^2, \quad Q = b + c, \quad P = b + 2c.$$

Let T be the triangle with sides (a, b, c) opposite (α, β, γ) . Thus $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha + \beta$. There exists a triangle with angles $(2\alpha, \alpha, 2\beta)$ tiled by N congruent copies of T if and only if

$$N = k^2 QP \quad \text{for some integer } k \geq 1.$$

In the units of the specified tile, the target sides are $k(c^2, cQ, bP)$. No condition $c < 2b$ is needed.

Construction

1. The known piece. Apply Beeson [1, Theorem 11] with

$$M = a, \quad K = c, \quad N_W = cQ.$$

Its hypotheses hold because $a^2 = c(c - b)$, hence

$$M^2 + N_W = 2K^2, \quad N_W - M^2 = 2bc > 0, \quad K \mid N_W.$$

The output tile is $(M, N_W/K - K, K) = (a, b, c)$. The theorem supplies a cQ -tiling of a triangle ABC with

$$AB = c^2, \quad BC = aQ, \quad CA = bc,$$

whose angles at A, B, C are $2\alpha, \beta, \theta = \alpha + \beta$, respectively. Only this sufficient construction is used.

2. The second piece and the join. Take a Q -fold similar copy of T ; its sides are aQ, bQ, cQ , and its standard quadratic subdivision contains Q^2 copies of T . Glue its aQ side to BC , with its interior in the opposite half-plane. Match its β endpoint to B and its γ endpoint to C ; call the remaining vertex H . At B the angles add to 2β , and at C

$$\theta + \gamma = (\alpha + \beta) + (2\alpha + \beta) = \pi.$$

Thus A, C, H are collinear, with C between A and H . The union is the convex triangle ABH , with angles $2\alpha, 2\beta, \alpha$. Its sides and count are

$$AB = c^2, \quad BH = cQ, \quad AH = bc + bQ = bP, \quad N = cQ + Q^2 = QP.$$

The pieces have disjoint interiors by their placement on opposite sides of BC . Individual subdivision vertices on BC need not match: T-junctions are allowed. For any integer $k \geq 1$, scale the entire construction by k and subdivide each enlarged tile into k^2 original tiles. This yields $k^2 QP$ tiles.

Why these are all possible counts for this tile and shape

The integer side triple (c^2, cQ, bP) is primitive. Indeed $\gcd(c, Q) = 1$ and $\gcd(c, bP) = 1$, because $\gcd(u, v) = 1$; hence

$$\gcd(c^2, cQ, bP) = \gcd(c, bP) = 1.$$

Any triangle of this shape has side lengths $s(c^2, cQ, bP)$ for some real $s > 0$. If it is tiled by T , each external side is a chain of whole edges of integer lengths a, b, c , so all three target lengths are integers. Choose integers x, y, z with $xc^2 + ycQ + zbP = 1$. Then

$$s = x(sc^2) + y(scQ) + z(sbP) \in \mathbb{Z}.$$

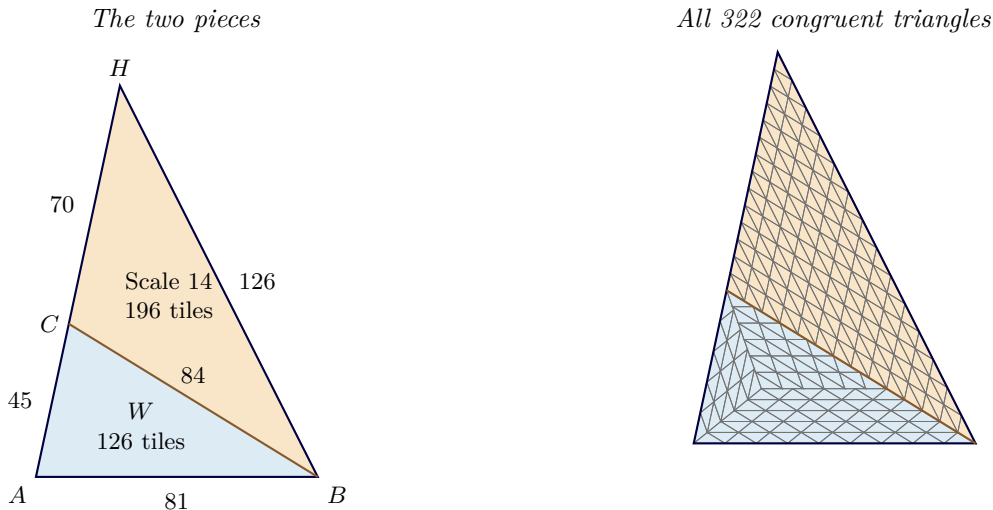
Thus $s = k \geq 1$ is an integer. Comparing areas with the construction above gives $N = k^2QP$, proving necessity. No coloring equation is required.

The explicit 322-tile example

For $(u, v) = (2, 3)$, we have $(a, b, c) = (6, 5, 9)$, $Q = 14$ and $P = 23$. The known piece has sides $(81, 84, 45)$ and 126 tiles; the second piece has sides $(84, 70, 126)$ and 196 tiles. Gluing their length-84 sides with the prescribed endpoint matching gives

$$(81, 115, 126) \text{ tiled by } 126 + 196 = 322 \text{ copies of } (6, 5, 9).$$

This is the 322-tile instance listed as unresolved in the remark following Theorem 14 of [1]. The known 126-tile piece is also illustrated in [1, Figure 14].



The drawing is generated from the explicit tile coordinates. In those coordinates (x, y) represents $(x, y\sqrt{32})$; the four vertices before a rigid motion are $A = (25, 0)$, $B = (-38, -9)$, $C = (-10, 5)$ and $H = (-580/9, 115/9)$. The supplied exact-coordinate dissection has been checked for congruence, containment, total area and pairwise disjoint interiors using rational arithmetic. This computation is supplementary to the general proof.

Reference

[1] Michael Beeson, *Triangle Tiling: The Case $3\alpha + 2\beta = \pi$* , arXiv:1206.2229v4, 25 September 2026. Theorem 11, Figure 14, and the remark following Theorem 14. <https://arxiv.org/abs/1206.2229v4>.